

**UNITED STATES OF AMERICA
BEFORE THE
FEDERAL ENERGY REGULATORY COMMISSION**

Interconnection of Large Loads to the Interstate) Docket No. RM26-4-000
Transmission System)

COMMENTS OF THE INDICATED PJM TRANSMISSION OWNERS

Pursuant to the Federal Energy Regulatory Commission’s (“Commission”) October 27, 2025, Notice Inviting Comments on the Secretary of Energy’s proposed advance notice of proposed rulemaking (“ANOPR”) requested under section 403 of the Department of Energy Reorganization Act¹ on October 23, 2025, the Indicated PJM Transmission Owners² respectfully submit the following comments.³

I. EXECUTIVE SUMMARY

In his letter accompanying the ANOPR, the Secretary of Energy emphasizes the need to connect large load customers, particularly data center customers, to the grid in a “timely, orderly, and non-discriminatory manner.”⁴ The Indicated PJM TOs appreciate the importance of timely

¹ 42 U.S.C. § 7173.

² This filing is prepared on behalf of the following PJM Interconnection, L.L.C. (“PJM”) Transmission Owners: The Dayton Power and Light Company d/b/a AES Ohio (“AES Ohio”); Duke Energy Corporation on behalf of its affiliates Duke Energy Ohio, Inc., Duke Energy Kentucky, Inc., and Duke Energy Business Services LLC; Exelon Corporation on behalf of its affiliates Atlantic City Electric Company, Baltimore Gas and Electric Company, Commonwealth Edison Company, Commonwealth Edison Company of Indiana, Inc., Delmarva Power & Light Company, PECO Energy Company (“PECO”), and Potomac Electric Power Company; FirstEnergy Service Company, on behalf of its affiliates American Transmission Systems, Incorporated, Jersey Central Power & Light Company, Mid-Atlantic Interstate Transmission LLC, The Potomac Edison Company, Monongahela Power Company, Keystone Appalachian Transmission Company, and Trans-Allegheny Interstate Line Company; PPL Electric Utilities Corporation; and Rockland Electric Company (together, the “Indicated PJM TOs”).

³ Several of the Indicated PJM TOs are separately submitting company specific comments and participating in other sets of comments, thereby, the Commission should consider this submission alongside and in addition to the other comments submitted by the companies participating herein.

⁴ Secretary of Energy’s Letter to the Commissioners of FERC, at 1 (Oct. 23, 2025) (“Secretary’s Letter”).

integration of data center customers. As transmission-owning utilities operating in a number of the most popular locations for data center development in the country, the Indicated PJM TOs are at the forefront of efforts to efficiently and cost-effectively interconnect data centers and similar large load customers. At the same time, the Indicated PJM TOs continue to invest in the transmission system, ensure system reliability, and protect other retail customers from unjustified cost consequences and other potential adverse outcomes. To that end, the Indicated PJM TOs have been working with PJM, state regulators, large load customers, and other stakeholders to develop mechanisms to meet the challenges associated with the enormous increase in data center-driven load demand. As a result, the Indicated PJM TOs have a unique perspective on what reforms are likely to work to accomplish the Secretary of Energy's goals and what will not work—at least in the diverse PJM region.

Based on their experience in 14 different jurisdictions, the Indicated PJM TOs agree that the Commission can and should play a role in ensuring the timely, orderly, and properly cost allocated interconnection of large data center loads. However, to not delay achievement of these goals, it is important that any Commission action be appropriately targeted and respect existing paradigms and precedent regarding cost-allocation, federal and state jurisdiction, and protecting reliability. In particular, the Commission should ensure that any rules it adopts will work in harmony with efforts already well underway to effectively integrate large load customers, inclusive of data center customers, while protecting existing retail customers.

In this spirit, the Indicated PJM TOs offer comments on a number of the principles proposed in the ANOPR, including the following:

- The Commission should limit the scope of any rule to data center facilities 200 MW or greater. A 20 MW threshold as tentatively proposed in the ANOPR will sweep in numerous load customers for which the ANOPR was not designed. A 20 MW threshold would work at cross purposes to the efficiency goals underlying the ANOPR, create

unnecessary obstacles for non-data center loads, and risk undermining the Administration's objective to increase domestic manufacturing.

- Costs of transmission network upgrades triggered by interconnection of large load customers should, by default, be rolled into transmission rates consistent with longstanding Commission precedent on the treatment of integrated transmission facilities, rather than directly assigned to individual customers. Rolled-in rate treatment will promote more efficient and timely large load interconnection by avoiding cost allocation and jurisdictional disputes, as well as ensuring that all load customers continue to contribute their fair share of the benefits of an integrated transmission network. Additionally, transmission utilities such as the Indicated PJM TOs have already been successfully collaborating with state regulators, electric distribution companies, load serving entities, and data center customers to implement mechanisms to protect other retail customers from unfairly subsidizing data center interconnections. The Commission should bolster those efforts with an even-handed, across-the-board wholesale cost allocation to give the states a clean slate upon which to make retail rate determinations.
- Regardless of how the Commission decides to allocate the costs of network upgrades, it must consider how the effect of its action or inaction affects all customers—not just large load customers. In addition to considering how the Commission's action in this proceeding may affect all customers, the Commission must account for the financial integrity of the electric utilities it regulates by ensuring that transmission owners are permitted to earn a reasonable return on such facilities. Failing to do so would chill necessary capital investment and threaten the goals of the ANOPR, as well as other Administration goals regarding the need for additional transmission infrastructure. It would be contrary to law and practically untenable for transmission owners to construct, operate, and maintain these facilities, while taking on the associated risks, without the facilities going into the transmission owners' rate base on which they are allowed to earn a reasonable return.
- Large load customers should not have an option to build transmission upgrades. Large load customers are not electric utilities, and do not possess the expertise and knowledge associated with developing and constructing transmission upgrades. Nor do they have a legal obligation to ensure reliability. The ANOPR's goal of greater efficiency can best be met by having transmission-owning utilities, which are statutorily required to provide a public service, continue to develop and to construct upgrades necessary to accommodate load customers.

II. COMMENTS

A. The Definition of “Large Loads” Must Be Appropriately Tailored to Meet the Goals of the ANOPR and to Avoid Adverse Consequences to Other Customers and U.S. Economic Priorities.

The ANOPR tentatively proposes that standardized interconnection procedures and agreements for large load should apply to loads of 20 MW or more, based on the threshold adopted by the Commission for distinguishing between “large” and “small” generators in the interconnection context. Generator interconnections and large load interconnections are not analogous, however. As applied to large load customers, a threshold of 20 MW is much too low given the ANOPR’s particular focus on data centers, and would both undermine the expressed goals of the ANOPR and impose significant costs and other burdens on traditional industrial and commercial load customers. The Indicated PJM TOs propose a two-tier threshold: (1) a MW threshold no lower than 200 MW, and (2) expressly limiting the application of any rule adopted by the Commission only to data center facilities, *i.e.* facilities that primarily utilize energy-consuming information equipment, as opposed to other types of large load customers.⁵

The ANOPR is primarily aimed at data center load, and the typical data center producing that load is far larger than 20 MW. Data centers contain energy-consuming information technology equipment that requires cooling and air management equipment—all of which use significant amounts of power. The tech industry suggests that the average data center contains up to 5,000 servers and that hyperscale data centers contain even more.⁶ According to a 2024 IBM

⁵ The Indicated PJM TOs use the terms data centers and large load interchangeably throughout this document. As discussed in this section, the Indicated PJM TOs believe that any large load interconnection procedures adopted by the Commission should be limited to data center customers. As such, references to “large load” should not be read to suggest that the Indicated PJM TOs concur that it is appropriate to apply standardized interconnection procedures to other types of load customers.

⁶ See IBM, *What is a hyperscale data center?*, <https://www.ibm.com/think/topics/hyperscale-data-center> (last visited, Nov. 18, 2025).

study, on average, 5,000 data center servers require approximately 100 MW of power capacity.⁷ Similarly, a 2024 Electric Power Research Institute report explained that new hyperscale data centers typically possess power capacities of 100 MW to 1,000 MW each, which is “roughly equivalent to the load from 80,000 to 800,000 homes.”⁸ Moreover, data centers have seen marked growth over the past decade-plus, getting larger and housing more power-intensive, higher-density servers. This has driven a massive increase in power consumption needs. While 10 years ago, a 20-30 MW data center was considered large, today 200 MW-plus facilities are typical.⁹

Also, the main driver of the increase in data center load is for AI-capable facilities, as these facilities must provide the high computational power and power density required by AI workloads.¹⁰ One recent analysis of projects found that, in a “midrange” scenario, AI-ready data center capacity needs will rise at an average rate of 33 percent a year between 2023 and 2030, with 70 percent of total demand for data center capacity by 2030 being for data centers equipped to host advanced-AI workloads.¹¹ This same analysis concludes that by 2030, 60 to 65 percent of AI workloads in both Europe and the United States will be hosted by so-called “hyperscalers” due to the capacity required to run large foundational models developed in-house, such as Google’s Gemini, or to host models developed by AI companies, such as OpenAI’s ChatGPT.¹²

⁷ *Id.*

⁸ EPRI, *Powering Intelligence: Analyzing Intelligence and Data Center Energy Consumption*, (May 28, 2024), <https://www.epri.com/research/products/3002028905> at 2.

⁹ McKinsey & Company, *AI power: Expanding data center capacity to meet growing demand*, (Oct. 29, 2024), <https://www.mckinsey.com/industries/technology-media-and-telecommunications/our-insights/ai-power-expanding-data-center-capacity-to-meet-growing-demand>.

¹⁰ *Id.*

¹¹ *Id.*

¹² *Id.*

In sum, typical data center loads already significantly exceed 20 MW and are anticipated to continue to increase substantially over the next several years. The Commission should choose a MW threshold value on the higher end of the current range of data center sizes so that it accounts for the trend towards larger-scale data centers.¹³ Based on these trends, the MW threshold for applying any Commission-adopted procedures relating to large load interconnections should be no less than 200 MW. Also, because large load additions are primarily made up of data center facilities, any rule adopted by the Commission should be limited to data center facilities. This distinction is not merely academic; it is critically important to properly define the scope of any Commission-adopted procedures at the outset both to achieve the goals of the ANOPR and to avoid unintended adverse consequences to other customers.

In the Secretary of Energy's letter accompanying the ANOPR, the Secretary points to the goal of driving American AI innovation, and the need to ensure that substantial transmission investment is done "efficiently, fairly and expeditiously."¹⁴ A threshold that is too low, and one that fails to distinguish between data centers and other load customers, will undermine these objectives. A 20 MW threshold would sweep in a multitude of load customers with a highly diverse set of characteristics, including manufacturing facilities, commercial businesses, and municipal load-serving entities. In many cases, the needs of these customers will not align with the needs of large data centers. Lumping all of these customers together runs the risk of creating new and unnecessarily large queues of customers, each of which will need to be processed, studied, and entered into appropriate agreements. One of the key lessons learned over the last twenty years

¹³ It would be less disruptive for the Commission to adopt a lower threshold in the future if, for some reason, 200 MW proves to be too high to accomplish the goals of the ANOPR, as opposed to adopting a threshold that is too low from the outset, which could lead to congested load request processing.

¹⁴ Secretary's Letter at 2.

is that large, clogged queues are anathema to fast and efficient interconnections.¹⁵ This concern is particularly acute in PJM, which has experienced exponential growth in data center loads in areas such as “data center alley” in Northern Virginia.

Moreover, different types of load customers have different operational attributes. Studying all of these types of customers in a single queue will create unnecessary complexity and reduce opportunities for streamlining and other efficiencies. Interconnection needs can also vary among data center customers, particularly with respect to smaller customers versus hyperscalers. A one-size-fits-all approach risks undermining the collaborative processes by which the Indicated PJM TOs have been able to work with customers of varying sizes and configurations to determine how best to interconnect new facilities. If the Commission wishes to promote timely interconnections, and advance AI innovation, it should limit the scope of any *pro forma* procedures to the facilities that will benefit most from those procedures both now and in the immediate future: large data centers. Otherwise, any procedures that the Commission adopts may end up simply creating more hurdles for both transmission owners and data center customers at a time when data center-related challenges are only going to increase for all participants.

A 20 MW or similarly low threshold that applies to all load customers, as opposed to being limited to data center facilities, also risks creating unnecessary impediments to non-data center loads, such as traditional industrial and commercial customers. In his letter to the Commission,

¹⁵ See, e.g., Mark Jaffe, *Grid operators across the country facing multi-year backlog in new generation connections*, EUCI.com (May 13, 2025), <https://www.euci.com/grid-operators-across-the-country-facing-multi-year-backlog-in-new-generation-connections/>; S&P Global, *Interconnection queues show swelling volume but FERC reforms slowly taking hold* (May 15, 2024), <https://www.spglobal.com/market-intelligence/en/news-insights/research/interconnection-queues-show-swelling-volume-but-ferc-reforms-slowly-taking-hold>; Lawrence Berkeley National Laboratory, *Grid connection backlog grows by 30% in 2023, dominated by requests for solar, wind, and energy storage* (Apr. 10, 2024), <https://emp.lbl.gov/news/grid-connection-backlog-grows-30-2023-dominated-requests-solar-wind-and-energy-storage>.

the Secretary of Energy notes that one of the Administration's priorities is revitalizing domestic manufacturing. The Indicated PJM TOs strongly support this goal. The large-scale buildout of the electric system has, for decades, driven innovation and helped create new types of industrial processes. This continues today, with manufacturing centers and other large load customers using more and more electricity.¹⁶ If the country is going to realize the challenge of preserving and expanding domestic manufacturing, these traditional load customers must not fall victim to discriminatory or other unreasonable treatment.

As noted above, the circumstances and needs of large data center customers are separate and apart from that of traditional large load customers. These other customers may, for example, have different tolerances for interconnection cost exposure or federal regulatory entanglements. Thus, the Indicated PJM TOs caution the Commission against applying procedures best suited for large data center customers to a broad swathe of other load customers.¹⁷ A broadly applicable set of new federal regulations, particularly ones that may involve significant up-front funding obligations, could end up deterring more traditional industrial and commercial customers from choosing the most efficient interconnection option, or worse, choosing not to locate in the United States at all. A significantly expanded interconnection process that sweeps in all load above 20 MW will also lead to delays and a lack of uniformity with respect to existing interconnection studies, undermining the efficacy of the current studies and procedures. Introducing variations in

¹⁶ U.S. Energy Industry Association Manufacturing Energy Consumption Survey, (Sept. 24, 2025), <https://www.eia.gov/consumption/manufacturing/>.

¹⁷ Some of these other large load power users include: (i) primary metal producers (iron, steel, and aluminum processes); (ii) chemical manufacturers (ammonia, chlorine, and petrochemical intermediaries); (iii) pulp and paper mills; (iv) petroleum refineries; (v) cement and glass manufacturers; (vi) transportation manufacturers including, large automotive manufacturing plants as well as aerospace products and parts manufacturing; (vii) year-round commercial indoor agricultural growing facilities; (viii) animal slaughterhouses and processing for meat consumption; and (ix) large hospitals and healthcare complexes.

the study process creates, at best, more opportunity for variations in the interconnection process, and at worst, incentives that undermine existing North American Electric Reliability Corporation (“NERC”) transmission planning performance standards.¹⁸

In addition to choosing an appropriate threshold, the Commission should also take into account data center load ramp periods. Many data centers’ initial loads are significantly lower when they first enter commercial operation, but following expansion and full potential of operation capacity, their demand increases to much higher levels. If the Commission elects to adopt standardized procedures for large load data center interconnections, it could consider applying the chosen MW threshold to loads with requested ramps that exceed the megawatt threshold at any point in the data center’s first five years of expected operation. This will ensure that data centers with relatively low initial load requirements when they enter service are not inappropriately exempted from applicable rules and procedures. This will also have the benefit of preventing strategic and potentially anti-competitive gaming of load ramp values.

B. The ANOPR’s Proposal to Require Allocating 100 Percent of Network Upgrade Costs to Large Load Customers Conflicts with the Commission’s Longstanding Cost Allocation Policies and Will Frustrate the ANOPR’s Goals of Efficient and Timely Interconnection.

Any mechanism for allocating the costs of network upgrades driven by large load customers must balance a number of objectives, including protecting existing retail load customers, encouraging the timely and efficient interconnection of data centers, promoting investment in necessary transmission infrastructure, and ensuring the financial integrity of the utilities that will be responsible for constructing that infrastructure. These objectives can best be achieved by the Commission recognizing rolled-in rate treatment as the default methodology for

¹⁸ NERC, Reliability Standard TPL-001-5.1 – Transmission System Planning Performance Requirements (June 13, 2025), <https://www.nerc.com/globalassets/standards/reliability-standards/tpl/tpl-001-5.1.pdf>.

allocating the costs of network upgrades, even those that are precipitated by new large loads.¹⁹ This approach will promote speed-to-power for data center customers by avoiding jurisdictional battles and other fights over cost allocation, and is fully consistent with robust protections for other load customers, which the Indicated PJM TOs have already been developing and implementing through tariffs and agreements for data center customers. Under these models, data center customers are responsible for the costs of upgrades necessary to reach the utility's system, including distribution system upgrades, as well as reasonable security and creditworthiness requirements. Finally, regardless of whether the Commission elects to adopt rolled-in rate treatment or a direct assignment model for large load-driven network upgrades, it must ensure that such upgrades are recoverable in the transmission-owning utilities' rate base on which they are permitted to earn a reasonable rate of return. Doing so is vital to attracting sufficient investment in necessary transmission infrastructure as well as protecting the financial viability of the utilities the Commission regulates.

1. Traditional Rolled-in Rate Treatment Best Accords with the ANOPR's Goals of Promoting Rapid Delivery of Vitrally Needed Data Centers.

Given that a chief concern giving rise to the ANOPR is to facilitate expeditious construction and activation of data centers to catch up to widespread demand, a rolled-in cost allocation rather than a targeted imposition of costs to discrete customers is the optimal approach in most cases.²⁰ Consistent with this objective, Commission policy has long favored rolled-in rate treatment for transmission network upgrades, absent special circumstances in which facilities are

¹⁹ This approach is not meant to preclude transmission owners from proposing direct assignment of network upgrades to data center customers through state jurisdictional retail tariff design or other mechanisms.

²⁰ Certain of the Indicated PJM TOs are actively exploring ways to mitigate retail load customer costs and are, in some cases, entering into arrangements that contain direct assignment constructs for certain customers.

not integrated with the grid.²¹ The Commission has found that costs applicable to integrated transmission facilities that are necessary for the interconnection of a load customer but that also benefit the network as a whole and all of its customers are appropriately rolled into transmission rates.²² Rolled-in rate treatment is the Commission's preferred cost allocation methodology for load-triggered network upgrades whether the load is large or small, and has consistently ruled that transmission facilities that are integrated into the transmission owner's network are eligible for rolled-in rate treatment.²³ These facilities benefit the entire transmission network and are not solely used by the large load customer. There is no reason for the Commission to deviate from this approach when it comes to determining the default methodology for allocating the costs of network upgrades triggered by the interconnection of data center customers or other large loads.²⁴ Doing so will result in renewed disputes and litigation over cost allocation, resulting in delay and, in turn, jeopardizing the goal of efficient and timely interconnection of large load customers.

The Commission has maintained a general policy preference for allocating transmission costs to all wholesale transmission customers rather than directly assigning those costs to individual customers, especially when the network upgrades benefit the broader grid or regional

²¹ See *Otter Tail Power Co.*, Opinion No. 93, 12 FERC ¶ 61,169, at 61,420 (1980); see also *Idaho Power Co.*, Opinion No. 13, 3 FERC ¶ 61,108 (1978); *Utah Power & Light Co.*, Opinion No. 220, 27 FERC ¶ 61,258, *reh'g denied*, 28 FERC ¶ 61,088 (1984), *aff'd sub nom. Sierra Pac. Power Co. v. FERC*, 793 F.2d 1086 (9th Cir. 1986); *Niagara Mohawk Power Corp.*, Opinion No. 296, 42 FERC ¶ 61,143 (1988).

²² See *Duke Energy Carolinas, LLC*, 168 FERC ¶ 61,190, at P 18 (2019) (citing *Ne. Tex. Elec. Coop.*, 108 FERC ¶ 61,084 (2004); *Am. Elec. Power Serv. Corp.*, 44 FERC ¶ 61,206 (1988), *reh'g denied*, 45 FERC ¶ 61,408 (1988), *reh'g denied*, 46 FERC ¶ 61,382 (1992)).

²³ *Allegheny Power*, 122 FERC ¶ 61,160 at P 29 (2008).

²⁴ The ANOPR confines its scope to interconnections "directly to transmission facilities" and makes no mention of local distribution-related upgrades or distribution rates. ANOPR at P 18. Nevertheless, the Commission should make explicit that nothing in this rulemaking proceeding is intended to regulate local distribution facilities or require distribution rate bypass for large load customers. Any other result would interfere with matters squarely reserved for state jurisdiction under the Federal Power Act. 16 U.S.C. § 824(b).

reliability. Over the years, the Commission has consistently found that transmission upgrades integrated with the larger network benefit the entire system, and therefore, network upgrade costs driven by large load customers should be rolled into transmission rates rather than directly assigned to a single customer.²⁵ This principle is rooted in the Commission's commitment to ensuring non-discriminatory access to the transmission system and promoting efficient grid development. As concerns about affordability continue to escalate, the importance of this principle only increases as it ensures that our national grid infrastructure is upgraded today in a cost-effective manner for not only the data centers of today but all load today and into the future.

The Commission has broadly construed its policy in favor of rolled-in rate treatment. For instance, the Commission has found that potential future conditions as well as present circumstances must be evaluated when determining whether a transmission facility is part of an integrated system.²⁶ The expansion of the transmission system to accommodate large load data center customers represent both the present and future transmission system needs the Commission foresaw in these cases when it sustained rolled-in rate treatment as the just and reasonable approach. The Commission has further held that "even remote facilities are part of the grid if they are merely the first step of what will eventually be a network loop," explaining that "an integrated transmission grid is a cohesive network moving energy in bulk that is in a dynamic state of development."²⁷

²⁵ See *Mansfield Mun. Elec. Dep't v. New England Power Co.*, 94 FERC ¶ 63,023, *aff'd*, Opinion No. 454, 97 FERC ¶ 61,134 (2001), *reh'g denied*, Opinion No. 454-A, 98 FERC ¶ 61,115 (2002).

²⁶ *Pinnacle West Capital Corp.*, 131 FERC ¶ 61,143, at P 42 (2010); *Allegheny Power*, 122 FERC ¶ 61,160, at P 23 (2008); *Utah Power & Light Co.*, 24 FERC ¶ 63,108, at 65,176 (1983); see also *Entergy Servs., Inc.*, 96 FERC ¶ 61,311, at 62,202 (2001).

²⁷ *S. Cal. Edison Co.*, 100 FERC ¶ 61,219, at P 18 (2002).

The Commission has also required rolled-in rate treatment for customers taking service from transmission facilities classified at different voltage levels. For instance, it has found that customers taking service from facilities operating at higher voltages also benefit from the transmission facilities operating at lower voltages.²⁸ In another proceeding, the Commission found that rolled-in rate treatment for the costs of transmission facilities that operate at lower voltages is appropriate where the lower-voltage facilities are integrated and support the operation of higher voltage facilities.²⁹ Moreover, the Commission has previously found that a transmission owner's 69-kV network in PJM is integrated with the networked transmission system, and thus, required the transmission owner to use rolled-in rate treatment for those facilities.³⁰

The Commission should retain the existing rolled-in rate approach as the default methodology for all load customers regardless of size, while recognizing the need for flexibility in individual circumstances.³¹ Expanding and maintaining the transmission network based on anticipated loads in a manner that benefits all load customers is the hallmark of prudent and effective transmission planning, and this is no less true when it comes to network upgrades triggered by data centers or other large load customers. Data centers or other large load customers are generally not the sole beneficiaries of network upgrades, even when the triggering cause of those upgrades is the interconnection of new loads. As the Commission has recognized, attempting

²⁸ *The Potomac Edison Co.*, 20 FERC ¶ 63,060 (1982), *aff'd*, Opinion No. 163, 23 FERC ¶ 61,106 (1983).

²⁹ *Utah Power & Light Co.*, Opinion No. 220, 27 FERC ¶ 61,258 (this case is especially applicable where lower-voltage facilities and higher-voltage facilities rely on the lower-voltage facilities when transmission outages occur).

³⁰ *Buckeye Power, Inc. v. Am. Transmission Sys. Inc.*, Opinion No. 533, 148 FERC ¶ 61,174 (2014).

³¹ The Indicated PJM TOs recognize that large load interconnections may also require customer-specific, non-integrated transmission upgrades, and that such upgrades may be appropriate for direct assignment to individual large load customers.

to parcel out the transmission network into discrete chunks allocated to specific customers fails to capture the benefits a cohesive, integrated system provides to all users.³²

Retaining the Commission's rolled-in rate policy ensures that large load customers pay for the benefits they receive from the integrated transmission system over the long term. The ANOPR does not demonstrate any special circumstances associated with large load customers, or otherwise provide a rationale justifying deviation from the Commission's rolled-in rate policy. Therefore, the just and reasonable approach is to determine that network upgrade costs driven by the interconnection of large load data center customers shall be allocated and shared amongst all load in accordance with the Commission's rolled-in rate policy.

Retaining the rolled-in rate policy will also promote the ANOPR's goals of efficient and timely interconnection. As the Commission is aware, issues involving the appropriate cost allocation of network upgrades have spawned significant controversy and litigation in the context of generator interconnection requests.³³ Moreover, generators and load customers are not the same. If the Commission were to copy and paste its Order No. 2003 cost allocation methodology used for generation facility interconnections as the model to be used by large load interconnections, it will inevitably lead to litigation over network upgrade cost allocation for years to come, particularly given that data centers and generators are different types of customers with different

³² *Ne. Tex. Elec. Coop., Inc.*, 108 FERC ¶ 61,084, at P 47.

³³ See e.g., *Nat'l Ass'n of Regul. Util. Comm'rs v. FERC*, 475 F.3d 1277 (D.C. Cir. 2007); *Midwest ISO Transmission Owners v. FERC*, 373 F.3d 1361 (D.C. Cir. 2004); *Midwest Indep. Transmission Sys. Operator, Inc.*, 129 FERC ¶ 61,060, at P 1 (2009); *Ameren Servs. Co. v. FERC*, 880 F.3d 571 (D.C. Cir. 2018); *Am. Clean Power Ass'n v. FERC*, 54 F.4th 722, 725 (D.C. Cir. 2022); *PPL Elec. Utils. Corp.*, 177 FERC ¶ 61,123 (2021); *Central Hudson Gas & Elec. Corp. v. N.Y. Indep. Sys. Operator, Inc.*, 176 FERC ¶ 61,149 (2021); *Central Hudson Gas & Elec. Corp. v. FERC*, 138 F.4th 531 (D.C. Cir. 2025); *Midcontinent Indep. Sys. Operator, Inc.*, 187 FERC ¶ 61,170 (2024); *Sw. Power Pool, Inc.*, 180 FERC ¶ 61,139 (2022); *Sw. Power Pool, Inc.*, 181 FERC ¶ 61,096 (2022); *Sw. Power Pool, Inc.*, 183 FERC ¶ 61,015 (2023); *RENEW Ne., Inc. v. ISO New England Inc.*, 189 FERC ¶ 61,216 (2024); *RWE Clean Energy, LLC v. PJM Interconnection, L.L.C.*, Complaint, Docket No. EL26-7-000 (filed Oct. 27, 2025).

needs and impacts on the transmission system. This will cast regulatory uncertainty over data center interconnections at precisely the time when predictability and certainty are most needed.

Conversely, incorporating large load data centers into the pool of customers paying for utility infrastructure that serves not just them, but all load, will ultimately drive rates down for all customers, and thereby make the national grid more affordable for all. The direct assignment of network upgrades may have appeal in the near term, but in the long term, it would effectively relieve large load from contributing to the continued development and maintenance of the transmission system, or any other part of the transmission system they are using. Over time, this means that large loads (that may even grow at a certain point of interconnection) would contribute less and less (as network upgrades depreciate) until they are contributing nothing at all to the network they continue to use.

However, if the Commission instead, consistent with its precedent, applies its tried and true rolled-in rate treatment policy as the default allocation for network upgrade costs driven by large load customers, it will provide certainty and stability for all customers and avoid unnecessary delays caused by cost allocation fights. Moreover, direct assignment of network upgrade costs to large load customers could encourage location shopping and speculative behavior, to the detriment of efficiency and fairness, and with further sacrifice owing to the resultant need for restudies. For these reasons, rolled-in rate treatment best harmonizes with the ANOPR's goals of fast, efficient, and fair interconnection of large load data center customers.

2. Transmission Owners and Electric Distribution Companies in PJM and Across the Nation Are Already Implementing Effective Mechanisms To Protect Other Load Customers from Impacts of Large Load Interconnections.

Recent experience across multiple jurisdictions demonstrates that utilities, system operators, state and federal regulators, and data center customers can, and have, worked together

effectively to design interconnection and rate mechanisms that uphold cost-causation principles while protecting other customers. Through collaborative tariff development, innovative contract structures, and coordinated state–utility review processes, these stakeholders have shown that rapid data center load growth can be accommodated without unfairly shifting the cost of new infrastructure to other retail customers.

The success of the models that are being developed provide a strong foundation for the Commission to build upon while preserving the flexibility to accommodate specific regional, state, and customer circumstances. On the other hand, requiring direct assignment to large load customers of all transmission upgrade costs associated with large load interconnections will undercut the significant progress that utilities and system operators throughout the country, including the Indicated PJM TOs, have made in working with state regulators and data center customers to implement mechanisms that balance cost allocation principles with appropriate protections to other retail customers.

One mechanism that has been successfully utilized is agreements between utilities and large load customers. These agreements typically incorporate a comprehensive package of terms designed to protect retail customers. Recent examples in PJM include the PECO Transmission Security Agreement (“TSA”) and the Dayton Power and Light Company d/b/a AES Ohio Construction Service Agreement (“CSA”).³⁴ Numerous other utilities are working to develop similar types of agreements.

These agreements take various forms, generally including provisions that allocate the costs and/or financial risks associated with the interconnection of and service to data center

³⁴ PECO Energy Company, TSA Filing, Docket No. ER25-3492-000 (filed Sept. 23, 2025); Dayton Power and Light Company d/b/a AES Ohio, CSA Filing, Docket No. ER25-192-000 (filed Oct. 23, 2024).

developments, including defined load and phasing schedules.³⁵ These schedules set the anticipated total load pursuant to which the data center is expected to operate. Such provisions protect existing retail customers against the risk of cost shifts if load development lags, scales down, or fails to enter service. These types of agreements also impose robust credit support obligations on the data center customer throughout the term of the agreement. For instance, under the PECO TSA, the customer must provide a parent guarantee or irrevocable letter of credit sufficient to cover its committed revenue contribution, with PECO authorized to draw on that security to recover unpaid costs.³⁶ The AES Ohio CSA likewise requires the customer to post a guaranty equal to the estimated network upgrade costs, declining proportionally each year for ten years after the commercial operation date.³⁷ These financial backstops increase the likelihood that electric infrastructure is developed only for viable projects, while providing rate stability and protections for other customers from exposure to project default or early termination. The agreements also establish a minimum or “committed revenue contribution” that the customer must pay annually, regardless of its actual usage. If the data center customer’s usage falls short of its anticipated usage (with some flexibility allowed), the customer must make shortfall payments equal to the difference between the committed and actual revenue contributions.³⁸

Regulators and utilities across multiple jurisdictions have also developed tariff-based approaches to ensuring that large load interconnections do not unfairly impact existing retail customers. These new tariffs share common features, including minimum-take or committed-revenue requirements, load development schedules, enhanced financial assurance measures,

³⁵ PECO TSA at Section 3 and Exhibit B; AES Ohio CSA at Section 3 and Exhibit B.

³⁶ PECO Transmittal at 5.

³⁷ AES Ohio Transmittal at 6.

³⁸ PECO Transmittal at 6-7, 10-11; AES Ohio Transmittal at 8-9.

shortfall/make-whole payments, early-termination charges, and clearer rules distinguishing distribution versus transmission cost responsibility.

The Public Utilities Commission of Ohio’s recent adoption of a data-center specific tariff (Schedule DCT) for AEP-Ohio illustrates this model. The approved tariff applies rigorous financial and minimum-usage conditions for new data center loads and is intended to lift a prior moratorium on new service while protecting existing customers from speculative project risk.³⁹ Indiana Michigan Power Company (“I&M”) is another example of a utility that has developed a data center-specific tariff. I&M’s settlement and tariff modification, approved by the Indiana Utility Regulatory Commission, revises its industrial/large-load pricing to reflect minimum demand thresholds, aggregated-site rules, and other protections negotiated with stakeholders.⁴⁰ It is important to preserve a flexible construct so that transmission owners and large load customers may enter into bilateral agreements like these on an individual case-by-case basis depending on the customer’s commercial needs. Adaptability to a customer’s unique needs is key in developing the appropriate menu of service. The Commission should avoid the temptation to craft a one-size-fits-all approach, which may limit the ability of transmission owners to best serve a variety of large load customers.

Additionally, PJM is undertaking efforts to prepare the grid for the influx of large load data center customers. On September 15, 2025, PJM initiated an accelerated Critical Issue Fast Path stakeholder process to determine solutions to resource adequacy challenges that are evident as more large loads plan to consume power from the system. PJM is considering several stakeholder

³⁹ AEP Ohio, Data Center Tariff, <https://www.aepohio.com/company/about/rates/data-center-tariff/> (last visited, Nov. 18, 2025).

⁴⁰ Indiana Michigan Power Company, *In the Matter of the Verified Petition of Indiana Michigan Power Company for Approval of Modifications to its Industrial Power Tariff – Tariff I.P.*, Cause No. 46097 (Indiana Util. Regul. Comm’n Feb. 19, 2025).

proposals that include the implementation of financial security requirements for large load customers to help address the capacity requirements of their facilities.⁴¹ This effort is highlighted in an October 17, 2025, letter sent by PJM’s CEO to Commissioner David Rosner.⁴²

The efforts discussed above demonstrate that utilities and large load customers can successfully implement arrangements that internalize risk with the project proponent and preserve fairness for existing customers, while ensuring that needed transmission and interconnection infrastructure is developed to allow data centers and other large load customers to obtain reliable power supply on a timely basis. Data center customers will, under these models, be responsible for the costs of facilities necessary to reach the utility’s system, as well as reasonable security and creditworthiness requirements, but will not be directly assigned the cost of upgrades to the integrated network. Moreover, these efforts will better promote speed-to-power compared to assigning the full costs of network upgrades to large load customers. This is because the service agreements described help reduce up-front financial commitments, and avoid any associated disputes, in favor of long-term service and demand obligations. They also are intended to reduce customer uncertainty about the magnitude of cost responsibility, a factor that in the context of generator interconnection has led to inefficiencies such as location shopping, speculative queue behavior, and withdrawals that ultimately require time-consuming restudies. In short, the goals of the ANOPR would best be served by allowing regulators, utilities, and large load customers to continue to develop and implement appropriate retail mechanisms to ensure adequate protections

⁴¹PJM Inside Lines, *PJM Discusses Large Load Planning at FERC Annual Reliability Technical Conference* (Oct. 23, 2025), <https://insidelines.pjm.com/pjm-discusses-large-load-planning-at-ferc-annual-reliability-technical-conference/>.

⁴² *Reliability Technical Conference*, Docket No. AD25-8-000, Pre-Technical Conference Remarks of PJM Vice President Jason Connell (Oct. 20, 2025) – Attached Oct. 17 Letter from PJM CEO, Manu Asthana to Chairman of FERC, David Rosner.

for retail customers rather than enforcing a one-size-fits-all cost allocation rule that clashes with longstanding Commission policy.

Regardless of how the Commission acts on this and other issues, during the pendency of this rulemaking, nothing should prevent utilities, state regulators, and data center/large load customers from continuing to develop and implement these and other mechanisms to interconnect data centers and other large load customers, nor discourage public utilities from making filings with the Commission to address these issues.⁴³

3. Mandating Direct Assignment of Network Upgrade Costs to Large Load Customers Can Result in Higher Overall Costs to Other Retail Customers.

As discussed above, it is not necessary to directly assign the costs of network upgrades associated with large load interconnections to protect other retail customers from unfairly subsidizing those large loads. Moreover, the direct assignment of such costs, as opposed to utilizing rolled-in rate treatment, may result in higher costs to retail customers under certain circumstances. This is because interconnecting data centers, particularly hyperscale facilities, adds significant amounts of additional load to the system. In the PJM region these loads, like all end-use customers, are ultimately responsible for paying the costs of network transmission upgrades reflected in zonal transmission rates based on their retail demand. Thus, including large load customers in the pool of customers that are ultimately allocated rolled-in rates for network upgrades can actually drive down transmission costs to other retail customers because they reduce the proportion of costs assigned to those other customers.

If, however, data center customers are directly assigned the costs of network upgrades, the result will almost certainly be higher overall costs to other load customers. This problem will

⁴³ ANOPR at P 32.

exacerbate as more data centers interconnect and the integrated transmission system is further fragmented. The result will be a balkanized network where an increasingly small percentage of transmission costs are recovered from all customers, including data center customers, on a rolled-in basis. This two-tier system would be neither fair nor efficient, particularly for existing retail customers. Large load customers, just like any other load customers, should bear their share of the full embedded costs of the system, recognizing that all customers enjoy the benefits associated with an integrated transmission system. Any other result would be unjust, unreasonable, and unduly discriminatory.

Altogether, the cost allocation proposal included in the ANOPR is unclear and undeveloped. That aside, the proposals set forth herein by the Indicated PJM TOs will protect customers now and into the future, if accepted by the Commission. As mentioned previously, there are certain circumstances, on a case-by-case basis when a public utility and customer may agree on a limited direct assignment approach for network upgrade costs.⁴⁴ That limited circumstance, however, should be the exception rather than the rule. It is paramount that the Commission declines to mandate transmission owners to directly assign the costs of network upgrades to a large load customer.

C. Transmission Owners Must Be Adequately Compensated for Owning and Operating Network Upgrades.

For all the reasons described above, the Commission should eschew requiring the direct assignment of network upgrades to large load customers in favor of continuing its longstanding policy in favor of rolled-in rate treatment as the default approach. The Indicated PJM TOs

⁴⁴ The Indicated PJM TOs support direct assignment treatment for the costs of interconnection facilities on the customer-side of the point of interconnection as those costs do not benefit the broader transmission network and thus are appropriately borne by the large load customer under cost-causation principles.

recognize that the ANOPR's cost allocation proposal is not yet developed. However, if the Commission pursues a direct assignment approach, and assuming *arguendo* that it has the authority to require such direct assignments, then any such rule must clearly specify that transmission owners will be permitted to earn a reasonable return on their investments for these network upgrades. The Commission must ensure rates and recovery mechanisms permit transmission owners to finance the work to connect large load data center customers on reasonable terms consistent with *Hope and Bluefield*. Without the opportunity to attract capital to invest in network upgrades necessary to electrifying large load data centers construction of transmission infrastructure to win the AI race will simply not occur.

1. Utility Returns Must Be Commensurate with Risk and Sufficient To Maintain Financial Integrity.

Under traditional cost-of-service ratemaking, regulated utilities invest capital to build transmission facilities to serve customers and assume the obligation to provide safe and reliable services to those customers at a reasonable cost. In exchange, utilities are provided an opportunity to recover their prudently incurred costs and to earn a reasonable return on their investment for their shareholders. A key feature of this regulatory compact is the equitable balancing of shareholders' and ratepayers' interests.

Longstanding precedent recognizes that transmission owners must be adequately compensated for owning and operating all transmission facilities, including network upgrades associated with large load data center interconnections. This is a requirement of the Constitution of the United States of America ("Constitution") that is captured in the Federal Power Act ("FPA")

and the cases interpreting the requirements of the FPA.⁴⁵ Under *Hope*, rates must, at a minimum, “enable the company to operate successfully, to maintain its financial integrity, to attract capital, and to compensate its investors for the risks assumed.”⁴⁶ In *Ameren*, the U.S. Court of Appeals for the District of Columbia expressed concern that transmission owners were required to own and operate network upgrades with no profit or compensation.⁴⁷ On remand, the Commission agreed, finding that network upgrades present risks and the affected transmission owners should be compensated for those risks.⁴⁸

The same reasoning should apply in this proceeding. If transmission owners are compelled to operate large expansions to the transmission system without a commensurate return, then their risk profile diverges materially from the proxy group underpinning the Commission’s return on equity (“ROE”) determinations. Absent safeguards, the allowed return would fall below the constitutional standard of comparability. Put another way, denying transmission utilities the ability to earn a return on a significant portion of their overall investment is contrary to the fundamental regulatory compact pursuant to which investor and consumer interests are to be balanced; the utility has an obligation to serve, and in turn receives a stable level of earnings in return.⁴⁹

⁴⁵ *FPC v. Hope Nat. Gas Co.*, 320 U.S. 591, 603 (1944) (“*Hope*”); *Ameren Servs. Co. v. FERC*, 880 F.3d 571, 581 (D.C. Cir. 2018) (“*Ameren*”); *Farmers Union Cent. Exch., Inc. v. FERC*, 734 F.2d 1486, 1515 (D.C. Cir. 1984).

⁴⁶ *Hope*, 320 U.S. at 605.

⁴⁷ *Ameren*, 880 F.3d at 583-85.

⁴⁸ *Midcontinent Indep. Sys. Operator, Inc.*, 164 FERC ¶ 61,158 at P 31 (2018), *reh’g order*, 169 FERC ¶ 61,233 at P 39 (2019).

⁴⁹ See *Jersey Cent. Power & Light Co. v. FERC*, 810 F.2d 1168, 1189 (D.C. Cir. 1987) (Starr, J., concurring) (indicating that as part of the regulatory compact, ratepayers are provided universal service and protection against monopolistic profits, and utility investors receive “a level of stability in earnings and value less likely to be attained in the unregulated or moderately regulated sector”); see also *Hope*, 320 U.S. at 603.

2. The Commission Should Not Create Policy that Exposes Transmission Owners to Asymmetric Risks with No Attendant Compensation.

Transmission owners face significant risks associated with owning, operating, and maintaining interconnection-related network upgrades, regardless of whether those upgrades relate to generator or load interconnections. However, the Commission has indicated that it may not be just and reasonable for transmission owners to elect to earn a return on generator-related network upgrades.⁵⁰ If the Commission were to extend such treatment to network upgrades to serve large load interconnections, it would likewise produce asymmetrical risk and, without the ability to earn a reasonable return on that investment, there would be no attendant compensation for transmission owners. The Indicated PJM TOs face both operational (*i.e.*, business) and financial risk associated with network upgrades for generator interconnection and have demonstrated through extensive comments in previous Commission proceedings that the risks from owning and operating network upgrades are both significant and growing.⁵¹

⁵⁰ See *Midcontinent Indep. Sys. Operator*, Order to Show Cause, 187 FERC ¶ 61,170 at P 1 (2024) (finding that existing ISO and RTO tariffs “appear to be unjust, unreasonable, and unduly discriminatory or preferential” because they include provisions for transmission owners to unilaterally elect to fund generator interconnection-related network upgrades and recover those costs through a charge that provides a return on and of such costs from interconnection customers).

⁵¹ For example, in support of their June 30, 2021 filing in Docket No. ER21-2282, the PJM TOs submitted the affidavit of Mr. David W. Weaver, Vice President of Transmission Strategy with Exelon Corporation, wherein Mr. Weaver discussed the fact that transmission owners bear significant risks associated with installing, owning and operating network upgrades, including (1) operational and safety risks, (2) reliability and cybersecurity compliance risks; (3) environmental risks; (4) weather and climate risks; and (5) outage coordination risks. See *PPL Elec. Utils. Corp.*, Docket No. ER21-2282-000, Affidavit of David W. Weaver, P.E., Vice President of Transmission Strategy of Exelon (filed June 30, 2021), at PP 7, 11-36.

The supporting testimony of Ms. Ellen Lapson similarly discussed how the PJM Transmission Owners face an increasing level of uncompensated risk as more network upgrades are added to the transmission system. Ms. Lapson’s testimony further explained how the growing number of network upgrades also creates financial risk—that is, because transmission owners receive no compensation for network upgrades, the increasing number of network upgrades causes a proportional reduction in the amount of compensation that the PJM Transmission Owners receive for their other transmission facilities. See *PPL Elec. Utils. Corp.*, Prepared Testimony of Ellen Lapson, CFA on Behalf of the PJM Transmission Owners, Docket No. ER21-2282, at 4-5 (filed Jan. 13, 2022).

The Indicated PJM TOs reiterate in this proceeding the evidence they have provided in other proceedings demonstrating the risks associated with owning and operating network upgrades for generator interconnection, should such a rate treatment become relevant for network upgrades to serve large load customers. These same arguments apply with equal force to large load interconnection-driven network upgrades. Put simply, adding new network upgrades makes the system larger and its operation more complex. Whether network upgrades are funded by the transmission owner or an interconnection customer, the network upgrade assets are operated and maintained by the transmission owner's personnel, who manage and maintain the rest of the transmission system and are subject to similar risks and obligations. Operational and safety risks are separate from and in addition to development risk. From a financial viewpoint, under such a rate treatment, there is no reward commensurate with the risk. The transmission owner is burdened by the obligation to bear operational risk and exposures, without any return related to network upgrade assets if they are not included in rate base. This forces the PJM TOs to operate portions of their network at zero ROE, which increases their overall financial risk.

The construction, ownership, operation, and maintenance of network upgrades also creates significant financial risk to the PJM TOs, whether those network upgrades are for generator customers or designed to serve large load.⁵² While PJM TOs already own and operate some generator interconnection related network upgrades with no compensation, the number of network upgrades that will be required to accommodate large load data center customer interconnections necessary to satisfy federal policy goals surrounding the expeditious development of artificial

⁵² See *PPL Elec. Utils. Corp.*, Initial Brief of the PJM Transmission Owners, Docket No. ER21-2282, at 16-18 (filed Jan. 13, 2022); *PPL Elec. Utils. Corp.*, PJM Tariff Revisions to Implement Transmission Owners' Funding of Network Upgrades, Docket No. ER21-2282, Transmittal Letter at 15-17 (filed June 30, 2021).

intelligence infrastructure is already significantly increasing, and will continue to do so into the foreseeable future. As the number of transmission upgrades for which a transmission owner does not earn a return grows, so does capital risk.⁵³

3. Transmission Owners Cannot be Forced to Construct and Operate Vast Portions of the Transmission System without Compensation.

In the 2024 Show Cause Order, the Commission suggested that no specific returns or compensation for network upgrades relating to generator interconnections are necessary because the Commission’s proxy group ROE methodology used to determine a transmission owner’s ROE should automatically account for the facilities, *i.e.*, “enterprise-wide ratemaking.” As open access service providers, transmission owners have a duty to meet the demands of their customers and to plan such construction in a transparent planning process pursuant to Order Nos. 888 and 889. Under the enterprise-wide risk theory, transmission owners would be obligated to devote their own limited resources to operate and maintain their facilities for which they will earn no return. Because under this theory, the compensation comes from other projects, there is an obvious mismatch between who benefits and who pays if this theory has any validity, which it does not. This mismatch will become increasingly egregious as more facilities are constructed to accommodate the massive influx of data center loads. As the PJM TOs argued in response to the 2024 Show Cause Order, this contention is legally flawed⁵⁴ and “would confer the Commission

⁵³ See *PPL Elec. Utils. Corp.*, Prepared Suppl. Aff. of Seabron Adamson on Behalf of the PJM Transmission Owners, Docket No. ER21-2282, at 5-6 (filed Sept. 20, 2021); see also *PPL Elec. Utils. Corp.*, Joint Affidavit of David Hunger and Seabron Adamson, Docket No. ER21- 2282, at 15-17 (filed June 30, 2021).

⁵⁴ See, e.g., *Jersey Cent. Power & Light*, 810 F.2d at 1180; *Duquesne Light Co. v. Barasch*, 488 U.S. 299, 307 (1989) (“*Duquesne Light Co.*”); *Bluefield Waterworks & Imp. Co. v. Pub. Serv. Comm’n of W. Va.*, 262 U.S. 679, 690 (1923) (“*Bluefield*”).

with limitless ratemaking authority to justify borrowing from Peter to pay Paul across a utility's enterprise."⁵⁵

Imposing enterprise-wide risk theory is also an impermissible taking under the Constitution and would impose confiscatory rates contrary to the Constitution and the FPA. As the U.S. Supreme Court has made clear, "the Constitution protects utilities from being limited to a charge for their property that is so 'unjust' as to be confiscatory."⁵⁶ Rates that "are not sufficient to yield a reasonable return on the value of the property used at the time it is being used [are] confiscatory, and their enforcement deprives the public utility company of its property."⁵⁷ In other words, transmission owners would be forced to operate a substantial part of their system on a non-profit basis.

Under the ANOPR's proposal, transmission-owning utilities would be required to own and operate a growing number of network upgrades without compensation. Compensation for a transmission owner occurs when rate base is multiplied by ROE. If assets are placed into rate base at \$0, there is no return. The Commission's ROE methodology does not contain any special procedures to produce compensation fine-tuned to assign a higher ROE to compensate for the risks of network upgrades not included in rate base.⁵⁸ Requiring no return on a portion of transmission assets places significant financial strain on utilities—network upgrades present the same risks as other transmission facilities, as discussed above.⁵⁹ The fact that a utility may earn other returns

⁵⁵ See Answer to Order to Show Cause of PJM Transmission Owners, Docket No. EL24-81-000, at 4 & n.8 (filed Sept. 11, 2024) ("Answer to Order to Show Cause").

⁵⁶ *Duquesne Light Co.*, 488 U.S. at 307.

⁵⁷ *Bluefield*, 262 U.S. at 690.

⁵⁸ Attachment 1 to Answer to Order to Show Cause, Prepared Testimony of Ellen Lapson, CFA, at 10-11 ("Lapson Testimony").

⁵⁹ Answer to Show Cause at 5.

from other parts of its business in no way ensures that the part of the business that gets nothing—*i.e.*, the business unit responsible for operating and maintaining network upgrades—is appropriately compensated.⁶⁰

Concerns regarding the risk associated with owning and operating load-triggered network upgrades are particularly acute here. The percentage of electricity consumption relating to data centers is expected to double (or roughly triple at the high end) by 2028—from roughly 4.4 percent in 2023 to up to 12 percent by 2028.⁶¹ New load growth related to data centers is estimated to account for roughly 30 to 60 percent of U.S. electricity load growth.⁶² It is untenable for transmission owners to be expected to construct, operate, and maintain the necessary facilities to accommodate this load growth on a not-for-profit basis.

Investing in and earning a return on facilities to accommodate customer load is the very core of the utility business. The inability to earn a stable and reasonable return on these facilities would send a chilling signal to the investment community and threaten the efficient and expeditious expansion of the transmission system necessary to revitalize domestic manufacturing and promote U.S. AI innovation. Moreover, if the main goal of the ANOPR is to more efficiently interconnect large load data center customers as electricity demand grows at an accelerated pace, not allowing utilities to earn a return on network upgrades runs counter to these goals.

⁶⁰ See Lapson Testimony at 12 (explaining how “lack of compensation for some assets will also cause the investment community to increase the cost of capital provided to the electric transmission sector”).

⁶¹ Lawrence Berkeley National Laboratory, *2024 United States Data Center Energy Usage Report* (Dec. 2024), <https://escholarship.org/uc/item/32d6m0d1>, at 5-6.

⁶² See McKinsey & Company, *How data centers and the energy sector can sate AI’s hunger for power* (Sept. 17, 2024), <https://www.mckinsey.com/industries/private-capital/our-insights/how-data-centers-and-the-energy-sector-can-sate-ais-hunger-for-power> (explaining that data center load may make up between 30 and 40 percent of all net new demand added until 2030); see also Boston Consulting Group, *Breaking Barriers to Data Center Growth* (Jan. 20, 2025), <https://www.bcg.com/publications/2025/breaking-barriers-data-center-growth>? (explaining that data center power demand will account for up to 60 percent of total load growth in the US from 2023 through 2030).

The Commission must ensure rates and recovery mechanisms permit transmission owners to finance the work on reasonable terms consistent with *Hope and Bluefield*. Without necessary capital investment, spurred on by reasonable returns on such investment, transmission infrastructure simply will not get built at the pace needed to ensure that the U.S. remains at the forefront of the AI revolution.

D. The Proposal to Require Direct Assignment of Network Upgrade Costs Raises Thorny Jurisdictional Issues that Are Best Avoided in the Interest of Promoting Efficient and Timely Data Center Interconnections.

The ANOPR’s principle that “load and hybrid facilities should be responsible for 100% of the network upgrades that they are assigned through the interconnection studies”⁶³ raises difficult jurisdictional issues, resolution of which would impede and delay the ANOPR’s ultimate goal of “ensur[ing] the timely and orderly interconnection of large loads.”⁶⁴ To avoid the need to resolve those issues, the Commission should use the long-standing approach of rolling the costs of load-driven network upgrades into transmission rates recovered from wholesale transmission customers (namely, load-serving entities (“LSEs”)), leaving to states to regulate how LSEs allocate the transmission costs among their retail customers (including large load customers).

In *New York v. FERC*, 535 U.S. 1 (2002), the Supreme Court recognized that the Commission’s exercise of authority over retail transmission service raises “serious” jurisdictional issues for unbundled retail transmissions,⁶⁵ and even more “complicated” jurisdictional issues for bundled retail transmissions.⁶⁶ These issues arise because, while “the FPA authorizes FERC’s jurisdiction over interstate transmissions, without regard to whether the transmissions are sold to

⁶³ ANOPR at P 25.

⁶⁴ *Id.* at P 1.

⁶⁵ *N.Y. v. FERC*, 535 U.S. 1 at 27.

⁶⁶ *Id.* at 28.

a reseller or directly to a consumer,”⁶⁷ the FPA preserves the states’ authority over (1) retail sales of energy,⁶⁸ and (2) facilities used for distribution of energy to retail customers,⁶⁹ and the rates charged for such distribution services.⁷⁰

The context in which the jurisdictional issues were raised in *New York v. FERC* was Order No. 888’s requirement of “nondiscriminatory open-access transmission services” “to encourage lower electricity rates by structuring an orderly transition to competitive bulk power markets.”⁷¹ The context here is different, and the proposed federal action is more intrusive on state jurisdiction: a federal requirement to assign 100 percent of transmission network upgrade costs to a particular end-use retail customer (the large load customer). This would eliminate the states’ traditional authority to regulate how the utility allocates its total transmission costs among various retail customer rate classes and would be beyond the authority granted to the Commission by the United States Congress.⁷²

⁶⁷ *Id.* at 20.

⁶⁸ *Id.* at 17, 28 (quoting 16 U.S.C. § 824(b)).

⁶⁹ *Id.* at 22-23 (quoting 16 U.S.C. § 824(b)).

⁷⁰ *Detroit Edison Co. v. FERC*, 334 F.3d 48, 54 (D.C. Cir. 2003); *see generally* 16 U.S.C. § 824(a) (“Federal regulation of . . . the transmission of electric energy in interstate commerce and the sale of such energy at wholesale in interstate commerce . . . extend[s] only to those matters which are not subject to regulation by the States.”).

⁷¹ *N. Y. v. FERC*, 535 U.S. at 10 (internal quotation marks and citation omitted).

⁷² *See, e.g., In the Matter of Proactive Planning for Upgraded Electric Grid Infrastructure*, Case No. 24-E-0364, 2025 WL 1704347, at *13 (N.Y.P.S.C. June 12, 2025) (“The National Grid surcharge recoveries shall be initially *allocated to service classes* based on using the appropriate transmission and demand allocators as contained in the utility’s most recent cost of service study.”) (emphasis added); *In the Matter of the Provision of Basic Generation Service, the Compliance Tariff Filing Reflecting Changes to Schedule 12 Charges in PJM Open Access Transmission Tariff – April 2021 Joint Filing*, BPU Docket No. EO21040730, Decision and Order Approving the Pass-Through of FERC Authorized Changes in Firm Transmission Service Related Charges, 2021 WL 2653467, at *3 (N.J. Bd. Reg. Comm’rs June 24, 2021) (“Each EDC is allocated the derived NITS charges and TECs from PJM based on the cost in each EDC’s transmission zone. Based upon the *allocation of the TEC Filings transmission costs among the service classes*, the monthly bill for a residential customer using 650 kilowatt-hours per month will change by approximately the following amounts (including Sales and Use Tax): an increase of \$2.27 or 1.76% for

The ANOPR's reliance on other authorities similarly raises difficult jurisdictional issues in the instant context of the ANOPR's proposed federal requirement to assign 100 percent of transmission network upgrade costs to the large load customer.

First, Order No. 2003's concern with open and non-discriminatory access for generators does not, as explained above, clearly support the more intrusive step of the Commission dictating what transmission costs should appear on a particular retail customer's bill.

Second, *FERC v. Electric Power Supply Association*, 577 U.S. 260 (2016) ("*EPSA*"), does not support the ANOPR's conclusion that "interconnection of large loads to the transmission system falls under a practice directly affecting Commission-jurisdictional wholesale electricity rates[.]"⁷³ At issue in *EPSA* was a Commission rule about how demand response bidders would bid in a wholesale auction; the Supreme Court held that this easily qualified as a rule or practice that affected the wholesale rate and fit within the Commission's authority.⁷⁴ On the other hand, the ANOPR's proposal here to directly act upon the rate design applicable to a retail customer's bill in the name of promoting large load interconnections would raise serious jurisdictional issues, regardless of any ultimate impact on the wholesale energy price.⁷⁵

Third and *fourth*, for the reasons explained above, the ANOPR's proposal intrudes on traditional state authority by dictating the amount of transmission costs that appear on a retail

ACE, a decrease of \$0.37 or 0.41% for JCP&L, an increase of \$0.36 or 0.30% for PSE&G, and an increase of \$0.73 or 0.64% for Rockland.") (emphasis added).

⁷³ ANOPR at P 14 & n.33 (citing *EPSA*).

⁷⁴ 577 U.S. at 279 ("Wholesale demand response, in short, is all about reducing wholesale rates; so too, then, the rules and practices that determine how those programs operate.").

⁷⁵ See *EPSA*, 577 U.S. at 280 ("Suppose, to take a far-fetched example, that the Commission issued a regulation compelling every consumer to buy a certain amount of electricity on the retail market. Such a rule would necessarily determine the load purchased on the wholesale market too, and thus would alter wholesale prices. But even given that ineluctable consequence, the regulation would exceed FERC's authority, as defined in § 824(b), because it specifies terms of sale at retail—which is a job for the States alone.").

customer's bill, and such intrusion is not supported by a "core [FPA] purpose."⁷⁶ It is one thing to require, per *New York v. FERC*, open and non-discriminatory access to the transmission system for generators and wholesale transmission and energy customers, including in the situation where the transmission will ultimately go to retail customers in an unbundled manner. It is another thing for the Commission to regulate how state regulators assign the recovery of transmission costs to different classes of retail customers.

To be clear, the Commission has jurisdiction to fix the rates charged by transmission providers to wholesale transmission customers, and—as the Indicated PJM TOs argue elsewhere in these comments—the Commission should use that jurisdiction to require that costs benefiting the transmission network be rolled into the transmission providers' rate base and recovered, with a reasonable rate of return, from all network transmission customers. But the Commission should stop there, and refrain from raising serious jurisdictional issues about federal jurisdiction to dictate what transmission costs should appear on a particular retail customer's (including a large load customer's) bill. That approach will avoid the delay in litigating and resolving the jurisdictional issues and, hence, will speed interconnection of large load customers consistent with the Department of Energy's overall goal.

E. The Commission Should Not Grant Large Load Customers an Option to Build.

The ANOPR states that, to the extent the interconnection customer is not the transmission owner, the interconnection customer shall be afforded the same (or equivalent) option to build as currently provided to generator interconnection customers.⁷⁷ Under the Commission's current *pro*

⁷⁶ ANOPR at P 16.

⁷⁷ ANOPR at P 26.

forma large generator interconnection agreement,⁷⁸ generator interconnection customers have the option to assume responsibility for the design, procurement and construction of Transmission Provider's Interconnection Facilities⁷⁹ and Stand Alone Network Upgrades.⁸⁰ The Commission should decline to extend this, or any other, option to build to load customers. Doing so would raise reliability issues and make the interconnection process less efficient.⁸¹

The Indicated PJM TOs continue to disagree that an option to build, even in the context of generator interconnections, provides any significant value in terms of cost savings and efficiency when weighed against the need to ensure that any upgrades are constructed in a safe and reliable manner. Regardless, the Commission's endorsement of a generator option to build does not support extending that option to large load customers.

While a number of large load customers are highly sophisticated commercial actors, there is no indication that these entities possess the relevant knowledge, expertise, or staffing sufficient to undertake the development and construction of electric transmission facilities. Unlike many of the entities that develop generators, load customers are not electric utilities, and therefore, the

⁷⁸ *Pro forma* LGIA, Article 5.1.3 (Option to Build).

⁷⁹ Transmission Provider's Interconnection Facilities are defined as all facilities and equipment owned, controlled, or operated by Transmission Provider from the Point of Change of Ownership to the Point of Interconnection as identified in Appendix A to the Standard Large Generator Interconnection Agreement, including any modifications, additions or upgrades to such facilities and equipment. Transmission Provider's Interconnection Facilities are sole use facilities and shall not include Distribution Upgrades, Stand Alone Network Upgrades or Network Upgrades. *Pro forma* LGIA, (Definitions).

⁸⁰ Stand Alone Network Upgrades are defined as Network Upgrades that are not part of an Affected System that Interconnection Customer may construct without affecting day-to-day operations of the Transmission System during their construction and the following conditions are met: (1) a Substation Network Upgrade must only be required for a single Interconnection Customer in the Cluster and no other Interconnection Customer in that Cluster is required to interconnect to the same Substation Network Upgrades, and (2) a System Network Upgrade must only be required for a single Interconnection Customer in the Cluster, as indicated under Transmission Provider's Proportional Impact Method. *Pro forma* LGIA, (Definitions).

⁸¹ This proposal also raises the same concerns as discussed above regarding the ability of transmission owners to earn a fair return on investment.

Commission has no jurisdiction to compel them to observe the requirements of FPA Section 215 and the concomitant reliability standards imposed by NERC. There is no expectation that large load customers will have some level of relevant technical knowledge on power interconnection and operations as generator owners because they are not in the business of constructing and/or operating electric facilities. It makes little difference whether some large load customers have access to capital. That ability does not render their qualifications sufficient to construct high voltage electric transmission facilities.

Giving large load customers the option to build is more likely to delay large load customer interconnections than to accelerate them. Large load customers will need to acquire adequate expertise before even starting the engineering, design, and construction process which includes skills at designing, planning, land acquisitions, and permitting and siting, and some of these functions may still need to be managed by the transmission-owning utility. For instance, if the transmission utility continues to manage state siting and permitting applications and resulting proceedings, it would be necessary to develop a complex web of interactions between the large load customer, the utility and regulators that will likely result in significant delays in siting and permitting proceedings. It will also increase costs to large load customers because they will also have to compensate the transmission owners for their administrative costs to review the facility design and confirm that the facilities meet the requirements for such facilities in the transmission owner's franchise area. Providing large load customers an option to build will therefore frustrate the ANOPR's goal of accelerating the interconnection of large loads. The most efficient and cost-effective approach is to have experienced utilities continue to build the facilities needed to interconnect large load customers.

Also, there are a limited number of qualified professional electrical engineers with expertise in designing and planning the transmission system. The industry has to competitively solicit talent from all over the country and the globe to provide these critical services to electric customers. If talent is going to be diverted to a new subset of customers just so they may exercise the option to build, it will strain and dilute this limited pool of qualified personnel. This will further inhibit the speed and efficiency of not just large load interconnections, but system planning generally, to the detriment of all customers.

Providing large load customers an option to build also raises reliability concerns. The network upgrades associated with large load interconnections will be integrated with the transmission system and will be used to serve all customers. Transmission-owning public utilities, which are legally obligated to protect system safety and reliability, have the appropriate incentive to build and properly maintain network facilities. As end-use customers, large load customers have no obligations to protect or maintain the reliability of the transmission system, including compliance with applicable NERC standards.⁸² Accordingly, existing electric utilities, using existing expertise, tools and processes, are best positioned to efficiently and cost-effectively construct necessary network transmission upgrades driven by large load interconnections.⁸³

⁸² See *e.g.*, NERC, Reliability Standard TPL-001-5.1 – Transmission System Planning Performance Requirements (June 13, 2025), <https://www.nerc.com/globalassets/standards/reliability-standards/tpl/tpl-001-5.1.pdf>. Although the ANOPR states that NERC should review its reliability standards to determine if new registration categories or modified standards are necessary, this directive provides no assurances that large load customers will be subject to any relevant NERC obligations in the future, nor if so, when. ANOPR at P 31.

⁸³ Even if a basis exists for granting large load customers some option to build, which the Indicated PJM TOs maintain there is not given the goals of the ANOPR, any such option would need to be subject to conditions to ensure that the applicable transmission upgrades are constructed safely and reliably. First, any option to build should be limited solely to transmission interconnection facilities and stand-alone network upgrades, such as the customer side of the substation where service drops from the transmission grid to customer-specific facilities. Under no circumstances should an option to build extend to existing, energized facilities. Similarly, large load customers should be required to strictly adhere to the design and materials/equipment standards of the interconnecting transmission owner. Failure to enforce consistent

An option to build also will hinder the ANOPR's goal to accelerate large load interconnections because it is likely to lead to disputes over what facilities qualify for the option. Even the generator *pro forma* LGIA recognizes this likelihood by including a specific provision to address such disputes. Under Article 5.1.3 of the *pro form* LGIA, both the transmission provider and the interconnection customer must agree as to what constitutes Stand Alone Network Upgrades and identify them. That section further provides for certain procedures to be followed if the transmission provider and interconnection customer disagree about whether a particular network upgrade is a Stand Alone Network Upgrade.⁸⁴

F. Technical and Implementation Issues

Certain of the principles proposed in the ANOPR, particularly principles 3 and 5, suggest that it would be appropriate, for purposes of studying the grid impacts of “hybrid” large load interconnection, to net the amount of on-site load and generation.⁸⁵ However, it is important to ensure that the full amount of a customer's on-site load and generation can be reliably interconnected. While it may make sense to coordinate studies, load and generation should be studied separately in the event the system needs to accommodate the full amount of the on-site load, or on-site generation, without the other. For instance, PJM has presented a hybrid interconnection option to the Commission that would permit large load interconnecting at the same

design standards would inject additional uncertainties into the planning process, complicating the operations, communications, and systems required for reliable operations and potentially degrading the reliability of the system. A large load customer should also be required to utilize contractors approved by the transmission utility, and the transmission utility must have oversight of the engineering design and construction process. Ultimately, the need to enforce these standards defeats the purpose of proposing an option to build in the first place: improving the efficiency and speed of large load interconnections. This leads full circle back to the fact that the best way to achieve the ANOPR's goal of timely large load interconnections is to have experienced transmission-owning utilities design and build the transmission facilities needed to interconnect large load customers.

⁸⁴ *Pro forma* LGIA, Article 5.1.3 (Option to Build).

⁸⁵ ANOPR at PP 20, 22.

point of interconnection as generation to be separately metered.⁸⁶ In this construct, large load is studied separately from generation and could be interconnected to the grid pursuant to existing utility load interconnection processes.⁸⁷

When a facility engages in bidirectional power flows, reliability requirements can only be met through increasingly complex protection schemes. However, simply adding protection schemes is not a panacea or a substitute for robust system planning. The unrestrained proliferation of protection schemes, such as Remedial Action Schemes (“RASs”) will lead to more complexity, adding layers of vulnerability to the grid. Many of these special protection schemes are meant to be temporary measures and should not be viewed as long-term solutions to the reliability issues associated with adding large loads to the bulk electric system. Least cost measures that ignore prime opportunities to build and expand the transmission system for future loads may result in system configurations that fail to deliver on customer needs and fall short of ensuring long term reliability. The Commission should refrain from adopting proposals that may undermine reliability standards or otherwise weaken the reliability of the grid.

The Indicated PJM TOs agree with principle 4 of the ANOPR proposal to the extent that it requires reasonable study deposits and readiness requirements for large load interconnections. These measures will help ensure that large load customers are sufficiently invested in the load interconnection process, which will deter speculative or duplicative requests. The Indicated PJM TOs also agree that a large load customer should be responsible for ancillary services based on the full amount of their load like all other customers.

⁸⁶ *PJM Interconnection, L.L.C., et al.*, Docket No. EL25-49-000, Answer of PJM Interconnection, L.L.C., at 11 (filed Mar. 24, 2025).

⁸⁷ *Id.*

Principle 7 of the ANOPR indicates that studies for large loads that agree to be curtailable, and hybrid facilities that agree to be curtailable and dispatchable, should be expedited.⁸⁸ Only fully developed processes that ensure reliability and incorporate the transmission provider's need to separately study large loads can reasonably effectuate Principle 7. For instance, a curtailment-related priority assumes that the large load customer is curtailing its full expected load amount. But if the large load customer would establish various levels of curtailment of its load, that would in fact increase the studies utilities would need to perform and the amount of time to perform them when compared to a traditional large load study request. Rather than mandating a shorter study period for curtailable customers, the Commission could instead consider providing a study priority for curtailable large load customers relative to other large load customers, but limited to those large load customers that make a binding and enforceable commitment to offer a meaningful level of curtailment, both with respect to the amount of load subject to curtailment and the curtailment period. The Commission should also provide more time to study the existing queue together with these customers. Otherwise, such a priority will simply incentivize gaming by offering *de minimis* levels of curtailment that provide little to no system benefit and could result in system reliability issues.

G. The Commission Should Ensure that this Rulemaking Process Complies with the Requirements of the Administrative Procedure Act.

The Secretary's October 23, 2025 ANOPR outlines high-level issues for the Commission's consideration and requests that the agency "build upon these principles and work expeditiously to initiate a rulemaking and ultimately issue a final rule April 30, 2026."⁸⁹ The labeling of the Secretary's proposal as an ANOPR (*i.e.*, *Advance* Notice of Proposed Rulemaking) indicates that

⁸⁸ ANOPR at 24.

⁸⁹ Secretary's Letter at 2.

it is intended to “allow[] FERC to solicit comments and gauge public interest *before* initiating a formal rulemaking process,” including comments that “suggest the agency not move forward with a rule.”⁹⁰

If the Commission initiates a formal rulemaking, a forthcoming notice of proposed rulemaking must include more fulsome details that allow for meaningful comments on the substance of the proposed agency action in order to pass muster under the Administrative Procedure Act (“APA”).⁹¹ This is particularly true for issues such as the ANOPR’s discussion of the need for “appropriate transition plans” to implement any proposed reforms (but without suggesting what those transition plans should be) and its statement that “NERC should review its reliability standards to determine if new registration categories or new or modified reliability standards are required to ensure reliability of the [bulk electric system]” (without further details for NERC to follow).⁹²

As noted in the Secretary’s Letter, the Commission is required under 42 U.S.C. § 7173(b) to “consider and take final action on any proposal made by the Secretary . . . in an expeditious manner in accordance with such reasonable time limits as may be set by the Secretary for the completion of action by the Commission on any such proposal.”⁹³ To constitute such a “reasonable time limit,” however, the limit imposed by the Secretary must allow for the required rulemaking

⁹⁰ See Rulemaking Explainer: Navigating the Rulemaking Process of the Federal Energy Regulatory Commission, available at <https://www.ferc.gov/OPP/rulemaking-explainer> (last updated March 10, 2025) .

⁹¹ See, e.g., 5 U.S.C. § 553(b)(3) (“The notice [of proposed rulemaking] shall include . . . either the terms or substance of the proposed rule or a description of the subjects and issues involved.”); *Nat’l Lifeline Ass’n v. FCC*, 921 F.3d 1102, 1115 (D.C. Cir. 2019) (“To meet the rulemaking requirements of section 553 of the APA, an agency must provide sufficient factual detail and rationale for the rule to permit interested parties to comment meaningfully. . . . For notice to be sufficient, the final rule must be a logical outgrowth of the proposed rule in the sense that the original notice must adequately frame the subjects for discussion.” (internal quotation marks and citations omitted)).

⁹² ANOPR at PP 30-31.

⁹³ See Secretary’s Letter at 2 n.5.

procedures set forth in the APA, including a sufficiently specific NOPR and opportunity for meaningful comment.

The Secretary's acknowledgement that the rulemaking process has not yet been initiated and the fact that an ANOPR is a precursor to the rulemaking process further support that any forthcoming rulemaking must adhere to typical rulemaking requirements. Thus, although the Secretary's Letter requests that the Commission take final action by April 30, 2026, if this time limit is inconsistent with the full rulemaking process, it is not "reasonable" within the meaning of 42 U.S.C. § 7173(b) and would need to be extended to accommodate those requirements.

III. CONCLUSION

The Indicated PJM TOs appreciate the opportunity to submit comments on the ANOPR, and respectfully request that the Commission act in accordance with these comments.

Respectfully submitted,

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November 21, 2025